

西方经济学

Part 1 Introduction

Lecture 1A GDP

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Supplement Readings

- (1) M9; S2.¹
- (2) 其他资料: [观看](#)²电影《一出好戏 (2018)》。
- (3) 其他资料: [GM] THE DATA OF MACROECONOMICS
- (4) 其他资料: [国家统计局](#), [国家数据](#). Bureau of Economic Analysis; BEA Glossary.
- (5) China Statistical Yearbooks:
[2025](#), [2024](#), [2023](#), [2022](#), [2021](#),
[2020](#), [2019](#), [2018](#), [2017](#), [2016](#), [2015](#), [2014](#), [2013](#), [2012](#), [2011](#),
[2010](#), [2009](#), [2008](#), [2007](#), [2006](#), [2005](#), [2004](#), [2003](#), [2002](#), [2001](#),
[2000](#), [1999](#).

¹M 指代马工程教材, S 指代课外阅读材料沈坤荣教程。

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- (1) 掌握 GDP 的核算方法，了解 GDP 指标的缺陷。
- (2) 熟悉国民收入账户中的常见指标。
- (3) 会查阅中国和美国的 GDP 数据。
- (4) 掌握三个恒等式。
- (5) 领会马工程教材精神。



Outline

1 Measuring the Value of Economic Activity

- Gross Domestic Product
- Circular Flow Diagram
- The Product Approach to GDP
- The Expenditure Approach to GDP
- The Income Approach to GDP
- Other Measures of Income
- The Circular Flow of Funds
- Notes for China's GDP
- What is Included and What is Not?

2 Three Identities

- The Income-Expenditure Identity
- The Current Account Identity
- The Saving-Investment Identity

3 Examples

4 马工程教材疑难重点



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Measuring the Value of Economic Activity

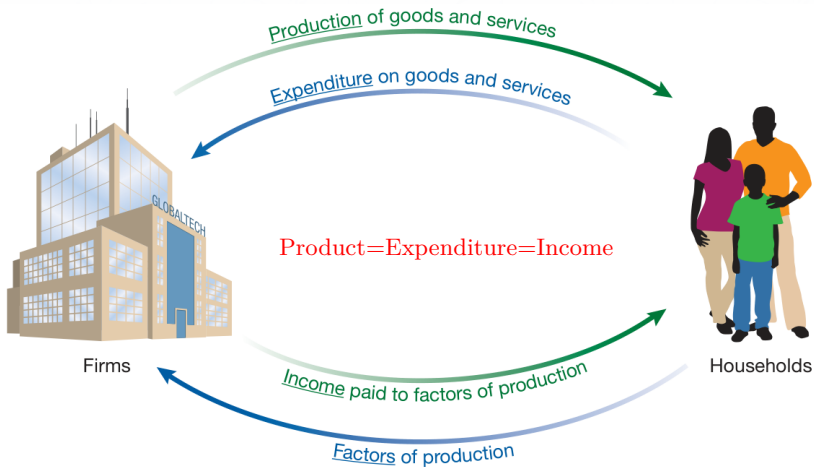
Gross Domestic Product

Definition 1

Gross domestic product (GDP) in National Income and Product Accounts (NIPA) is the market value of all final goods and services produced in an economy during a specified period, such as a year.

- (1) In a market;
- (2) It is a value such that different things can be added. Thus it can be affected by price changes;
- (3) Statistical scope: all, goods and services;
- (4) Final goods, rather than intermediate goods. Services are automatically final;
- (5) Produce. Transfer of used goods is not considered;
- (6) Regional scope: in an economy;
- (7) A flow variable: during a specified period.

Circular Flow Diagram: Acemoglu's (2022) Exhibit 5.1



The Product Approach to GDP

表 2-2 汽车生产过程的几个环节

	售价(万元)	中间产品价格(万元)	增值(万元)
铁矿石、煤炭等	10	0	10
钢铁	15	10	5
汽车	20	15	5
总计	45	25	20

The product approach is also called the *value-added* approach. The main principle in this approach is that GDP is calculated as the sum of value added to goods and services across all productive units in the economy.

Total Value Added

$$= \sum_i \left(\begin{aligned} &\text{Market value of goods and services produced by firm } i \\ &- \text{Market value of intermediate goods used in production by firm } i \end{aligned} \right)$$

The Expenditure Approach to GDP

Total Expenditure = $C + I + G + NX$.

- (1) C denotes **households' consumption** which is defined as the sum of final goods and services purchased by persons resident in the nation. It is disaggregated into three components: nondurable goods, durable goods, and services.
- (2) I denotes **gross private domestic investment** which is disaggregated into three components: nonresidential fixed investment (the purchase by firms of new plant and equipment), residential fixed investment (the purchase by households and landlords of new housing), and inventory investment.
- (3) G denotes **government purchases** of goods and services, which consist of purchases of consumption goods (G^C) and purchases of capital goods (G^I). Government expenditure is not exactly the government spending or outlays. Government spending includes not only purchases of goods and services, but also "transfer" payments and net interest payments. Actually, transfer and net interest payments do not involve new production. Thus transfer and net interest payments are not a part of GDP.
- (4) NX denotes **net exports** ($NX = \text{exports} - \text{imports}$), which is also called the **trade balance**. When $NX < 0$, we say a **trade deficit**. When $NX > 0$, a **trade surplus**. When $NX = 0$, a **balanced trade**. To calculate GDP, imported final goods and imported intermediate goods, already counted as part of C , I , or G , should be subtracted.

The Income Approach to GDP

We add up all income received by economic agents contributing to production.

$$\begin{aligned}\text{GDP} = & \text{Labor Income} + \text{Capital Income} \\ & + (\text{Indirect Business Taxes} - \text{Subsidies}) \\ & + \text{Depreciation}\end{aligned}$$

Capital income comprises rental income, corporate profits, net interest, and the unknown capital part of proprietors' income. Depreciation is also called consumption of fixed capital.



Other Measures of Income: GNP, NNP, NI

Gross national product (GNP) is the total income earned by a nation's permanent residents (called nationals). It is equal to GDP plus **net factor payments**(NFP) from abroad.

$$GNP = GDP + NFP,$$

$$NFP = \text{Factor Payments from Abroad} - \text{Factor Payments to Abroad}.$$

By comparison, GDP measures the total income produced *domestically*.
GNP is the total income earned by *nationals or residents of a nation*.

$$\text{Net national product}(NNP) = GNP - \text{Depreciation};$$

$$\text{Net domestic product}(NDP) = GDP - \text{Depreciation};$$

$$\text{National income}(NI) = NNP - \text{Statistical Discrepancy}.$$

$$\text{Gross} \leftrightarrow \text{Net} + \text{Depreciation};$$

$$\text{National} \leftrightarrow \text{Domestic} + NFP$$

Other Measures of Income: NI

National income (NI) is the sum of eight components ([NIPA Handbook](#), ch.2):

- (1) **Indirect business taxes** are called now **taxes on production and imports**. They include taxes on sales, excise, and value-added. **Subsidies**, if there are, enter into this category as a negative term since they are negative taxes. (Nonfactor charges)
- (2) **Business transfers** to persons, government, or the rest of the world, such as liability payments for personal injury, and corporate contributions to nonprofit organization. (Nonfactor charges)
- (3) **Current surplus of government enterprises**. (Nonfactor charges)
- (4) **Proprietors' income** is received by self-employed persons, including noncorporate businesses. Proprietors' income is a mix of payments to labor and capital, whose breakdown into labor and capital is unknown. (Factor income)
- (5) **Rental income of persons** is the income landlords receive, including imputed rent that homeowners pay to themselves, less expenses (e.g. depreciation). (Factor income)
- (6) **Net interest**: Interest paid by domestic private enterprises less interest received by domestic private enterprises, plus interest paid by the rest of the world less interest received by the rest of the world. (Factor income)
- (7) **Corporate profits** are the sum of taxes on corporate income, dividends and undistributed profits.³ (Factor income)
- (8) **Compensation of employees**, or labor income, which comprises wages, salaries, and fringe benefits earned by workers. Social insurance contributions are part of fringe benefits. (Factor income)

³If a firm's investment is financed by profits, this part belongs to undistributed profits while the investment enters into gross private domestic investment expenditure.

Other Measures of Income: PI, DPI

Personal income (PI) is the income that persons receive.

$PI = NI - \text{Indirect business taxes}$

- **Business transfers** + Personal current transfer receipts from gov and business
- **Current surplus of government enterprises**
- **Net interest** + Personal interest income
- **Corporate profits** + Personal dividends income
- Social insurance contribution (domestic)

$PI = \text{Personal current transfer receipts from gov and business}$

- + **Proprietors' income** + **Rental income of persons**
- + Personal interest income + Personal dividends income
- + **Compensation of employees** - Social insurance contribution (domestic).

Disposable Personal Income(DPI) is the amount households and noncorporate businesses have available to spend or save after satisfying their tax obligations to the government.

$DPI = PI - \text{Personal Current Taxes.}$



Other Measures of Income: NNFI and GNFI

Net national factor income (NNFI on line 32)

= National income – nonfactor charges

= National income

– Net Indirect business taxes

– **B**usiness transfers

– **S**urplus of government firms

= factor charges

= **P**roprietors' income

+ **R**ental income of persons

+ Net **I**nterest

+ **C**orporate profits

+ Compensation of **E**mployees

Gross national factor income (GNFI on line 29)

= NNFI + Consumption of fixed capital.



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早期国民收入概念

在上述国民生产净值中,如销售税之类的间接税是由企业交给政府的,构成消费者所支付的产品价格与企业所得到的收入之间的差额。间接税不构成企业的收入,所以不包括在国民收入之内。

$$NI = NNP - \text{间接税}$$

(四) 个人收入(PI)与个人可支配收入(PDI)

国民收入衡量了一国要素所有者的收入,但是国民收入并不等于个

马工程观点 (M, 2011, p.10)

国民收入(简称NI)是指一国全部生产要素在一定时期内提供服务所获得的报酬的总和,即工资、利息、租金和利润的总和。国民收入与国民生产净值的关系为:

$$NI = NNP - \text{间接税} - \text{企业转移支付} + \text{政府对企业的补贴}$$

所谓间接税是指税收负担不由纳税人本人承担的税,即这种税收的负担可以转嫁出去。企业馈赠的礼品是企业转移支付的例子。国民收入这一指标衡量经济中的所有人赚到了多少钱。

个人收入(简称PI)是指个人得到的收入。国民收入不是个人收入。

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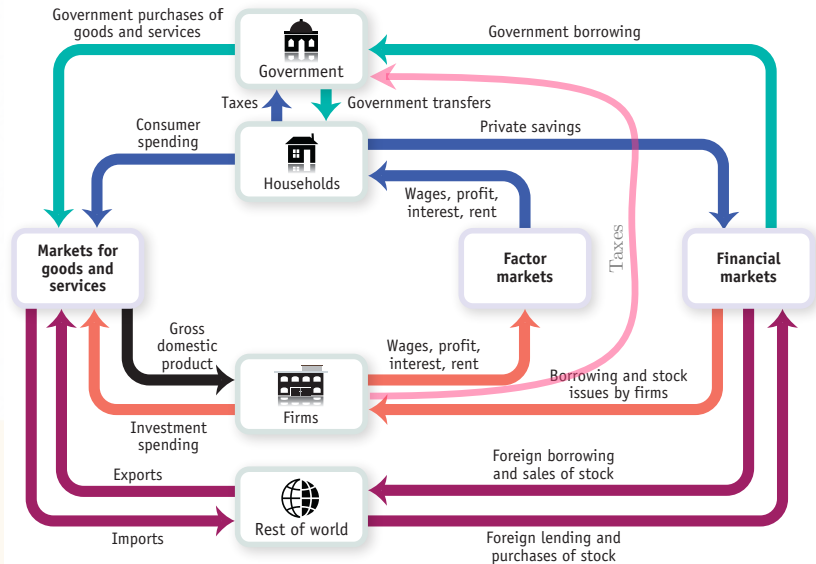
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The Circular Flow of Funds: Krugman and Weil (2015)



- (1) Aggregation inconsistency. The sum of provincial GDPs is greater than national GDP. (see [XinhuaNet](#))
- (2) Most of services are not covered in GDP until Dec., 2005. After the adjustment of scope, GDP in 2004 with adjustment increases by 16.8% relative to GDP in 2004 without adjustment. The adjustment is based on China economic census in 2004.
- (3) Traditionally regarded as intermediate inputs, Research and development expenditure entered GDP formally as formation of fixed investment in 2016.⁴

⁴2016 年 7 月 5 日, 国家统计局发布《[关于改革研发支出核算方法修订国内生产总值核算数据的公告](#)》。以前研发支出 (R&D) 作为中间消耗不计入 GDP。现在研发支出作为固定资本形成 (*I*) 计入 GDP。根据新的核算方法, 国家统计局修订了 1952 年以来的 GDP (国内生产总值) 数据。

What is Included and What is Not

- (1) **Market-price rule.** GDP includes only goods and services that transact in markets.
- (2) Exceptions to market-price rule, and imputation. An exception is **owner-occupied housing**. If you rent your house, the rental income is included in GDP. If you live in your house, national accountants estimate a rental equivalent which is included in GDP. The imputed rent is included both in the houseowner's expenditure and in the houseowner's income. Another exception is **government services** which are valued at cost since most of government services are not sold in a market.
- (3) No imputation is made for home production and the underground economy.
- (4) GDP omits physical capital depreciation and the **health** of a nation's people.
- (5) GDP does not include changes in **environmental resources**.
- (6) GDP does not consider income **distribution**.
- (7) GDP assigns no value to **leisure time**.
- (8) The sales of **used goods** is not included in GDP since it only reflects the transfer of an asset rather than an addition to the economy's income. Sales of used goods lead to no changes of income, no changes of expenditure, and no changes of production.
- (9) Investment in **inventory** is counted as an expenditure by producers. Thus production for inventory increases GDP just as much as production for final sale. A sale out of inventory is similar to the sale out of used goods and hence it does not affect GDP.

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The Income-Expenditure Identity

Proposition 2 (The Income-Expenditure Identity)

Let Y be the total income from domestic production. Then the following income-expenditure identity holds:

$$Y = C + I + G + NX. \quad (1)$$

Let $CA = NX + NFP$ denote **current-account balance**.

$$Y + NFP = C + I + G + CA,$$

CA measures the difference between total income (GNP) and domestic total expenditure ($C + I + G$, also called *domestic absorption*).



The Current Account Identity

Proposition 3 (The Current Account Identity)

The current account identity is given by

$$CA = S - \tilde{I}, \quad (2)$$

where S denotes national saving, which is the sum of private sector saving (S^{Prv}) and public sector saving (S^{Gov}). $\tilde{I}$ denotes national investment, which is the sum of private sector investment (I) and public sector investment (G^I). $S - \tilde{I}$ represents net foreign investment (NFI) or net capital outflow (CF). The current account identity is also called the fundamental balance of payments identity.

$$S^{Prv} = \text{Private sector disposable income} (Y^{Dis}) - \text{Private consumption} (C).$$

$$= [GNP - (T - TR - INT^G)] - C.^1$$

$$S^{Gov} = (T - TR - INT^G) - G^C.$$

$$S = S^{Prv} + S^{Gov} = Y + NFP - C - G^C = \tilde{I} + CA.$$

¹ Y^{Dis} is disposable income of the private sector. T is tax revenue. TR is government transfer payments. INT^G is net interest payments from government. If $NFP = 0$, $TR = 0$, and $INT^G = 0$, then $Y^{Dis} = Y - T$.

The Saving-Investment Identity

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Proposition 4 (The Saving-Investment Identity)

In a closed economy, the current account identity becomes saving-investment identity:

$$S = \tilde{I}. \quad (3)$$



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Example 1

观看电影《一出好戏》。假设所有交易都公开。孤岛上，马进和小兴面临饿死的绝境。一天，从天而降一大群鱼，拯救了马进和小兴。两人用天降之鱼交换到孤岛上的绝大多数物品。

- (a) 天降之鱼是否具有“使用价值”（依马克思定义）？
- (b) 天降之鱼是否具有“价值”（依马克思定义）？
- (c) 天降之鱼是否计入孤岛的 GDP？



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Example 2 (GDP 之谜)

有没有“神奇方法”创造 GDP?



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疑难重点

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- (1) 掌握 GDP 的概念，理解经济的资金环流图 (The circular flow of funds)。
- (2) 掌握 GDP 核算的三种方法：生产法、支出法、收入法。
- (3) 能够理解中国和美国的与 GDP 相关的经济统计表格。
- (4) 掌握下列指标之间的关系：GDP, GNP, NNP, NDP, NI, PI, DPI, GNFI, NNFI。
- (5) DPI 与 Private sector disposable income 有什么关系？
- (6) GDP 作为衡量经济活动的指标，有何缺陷？
- (7) 三个恒等式。



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1 (E2, p.27)

根据马工程教材观点，应当如何评析西方经济学的 GDP 指标？



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